Verified Localized Episodic Grounding: Robust Memory Consolidation for Large Language Models in Heterogeneous Environments

Abstract

Verified Localized Episodic Grounding (VLEG) combines local reflection generation, verification against labeled historical episodes, and retrieval of reflections with source evidence. Five symbolic experiments examine consolidation, grounding and refinement at 7B and 70B model scales. An additional public-flow experiment evaluates five historical-memory seeds on the same 1,000 independently partitioned, deduplicated UNSW-NB15 records using a 70B model. VLEG improves Macro-F1 over unverified first reflections by 17.02 percentage points, but underperforms raw flow retrieval: 59.68% versus 63.92%. Attack recall increases from 71.39% to 77.74%, accompanied by a false positive rate increase from 38.13% to 48.78%. The paired conditional 95% interval for the Macro-F1 difference is $[-5.91, -2.53]$ points. These results distinguish source-label checking and memory coverage from reliable downstream generalization. They support neither a universal hybrid-memory advantage nor operational intrusion-detection readiness. Historical symbolic numbers remain subject to provenance reconciliation; the completed flow experiment records exact model and source versions, prediction-level checkpoints and independently recomputed metrics.

Index Terms

Large Language Models, Episodic Memory, Semantic Abstraction, In-Context Learning, Hallucination Mitigation, Hybrid Grounding, Iterative Refinement

I. Introduction

A. Background: Retrieved Examples and Memory Consolidation

Large language models can use retrieved examples to support classification and analysis without parameter updates [1], [2]. In network analysis, such examples could be labeled historical flow records or incident summaries. Their usefulness depends on retaining relevant context while avoiding unsupported conclusions. This paper studies that memory-design problem using controlled symbolic tasks and a separate public-flow benchmark; it does not establish performance on operational production networks.

Persistent memory separates stored experience from the model's finite prompt window [3], [4], [5]. Raw-example retrieval preserves evidence, while semantic consolidation compresses episodes into summaries or heuristics [6], [7]. Compression can reduce repeated reasoning but can also remove exceptions or introduce claims that the evidence does not support.

VLEG combines local reflection generation, verification against the originating labeled episode, and retrieval of the reflection with its raw evidence. We evaluate whether these operations improve label consistency, memory coverage and downstream accuracy, and report the circumstances in which they do not. Network-flow analysis is evaluated in a separate public benchmark experiment with frozen historical-memory seeds and held-out flow fingerprints.

B. Problem Definition: Evidence and Context in Memory

Raw examples can burden inference as the number and complexity of episodes increase [8], [9]. Local reflections offer compact guidance, but a reflection can contradict its source label or cite irrelevant attributes. Feedback-based systems such as Reflexion and Voyager already use evaluation to improve later behavior [10], [11]; our focus is the narrower decision of whether to admit an individual reflection into memory after checking it against its source episode.

A global summary can represent context-dependent rules, including the background-conditioned rule used in our experiments. Heterogeneity alone does not make global consolidation potentially difficult under a restricted summary budget. The practical question is whether summarization preserves relevant conditions and exceptions within the available evidence and context budget [12], [13]. Local and global memory therefore require empirical comparison rather than an assumed ordering.

C. Verified Localized Episodic Grounding

VLEG generates a candidate reflection from an episode's attributes and checks it against the complete historical episode, including its known label. Rejected candidates receive a critique and can be revised up to a fixed budget K. Only accepted candidates are stored. The generator and verifier are separate calls to the same underlying model, not statistically independent models.

At inference, VLEG retrieves memory tuples and provides the model with both raw episodes and their accepted reflections. This hybrid context may preserve evidence while supplying a compact heuristic, but its benefit is an empirical question. The five symbolic experiments compare memory strategies, grounding modalities, iteration budgets and reflection diagnostics at 7B

and 70B model scales; a sixth experiment evaluates a controlled adaptation on public flow records. They show improved label consistency and memory coverage, alongside mixed downstream results and persistent unsupported attribution. These results establish boundaries of the evaluated implementation rather than universal correctness guarantees.

D. Paper Organization

Section II reviews related memory methods. Section III defines VLEG. Section IV reports the symbolic experiments, reports the additional public-flow benchmark, and discusses limitations. Section V concludes.

II. Related Work

A. Pure Episodic Retention and Direct Retrieval

The most fundamental paradigm in memory-augmented Large Language Models (LLMs) relies on Pure Episodic Retention, wherein the system stores raw, unmodified multi-modal or textual experiences. Within our proposed MECE framework, these methodologies are characterized by Single-Step Direct Generation (as no active abstraction occurs prior to inference) and Episodic-Only Retrieval (where the model is conditioned exclusively on raw historical samples).

The foundational instantiation of this approach is In-Context Learning (ICL), introduced by Brown et al. [1]. ICL demonstrated that LLMs can adapt to novel tasks simply by being provided with a few raw episodic examples in their prompt context, effectively utilizing the context window as a short-term working memory. While ICL preserves the high-fidelity details of each episode, its reliance on unstructured data severely limits its scalability. To address the inherent constraints of the context window, Lewis et al. [2] introduced Retrieval-Augmented Generation (RAG). RAG equips the LLM with a passive, long-term episodic datastore, dynamically retrieving the most relevant raw episodes at inference time.

Subsequent research within this paradigm has focused heavily on optimizing the retrieval and selection mechanisms to maximize the utility of the retrieved episodes. For instance, Zhang et al. [14] proposed active retrieval strategies to dynamically select the most informative examples for ICL, while Purohit et al. [15] explored sample-efficient retrieval techniques to minimize context bloat. Despite these optimizations, these approaches remain fundamentally constrained by their lack of an active abstraction mechanism. Because they rely on Single-Step Direct Generation over raw data at test time, the burden of inducing latent rules from a noisy sea of unstructured experiences falls entirely on the inference model's ad-hoc reasoning capabilities.

Retrieval quality and context selection affect how useful historical examples are [8]. Small symbolic tasks and flow records can expose different failure modes, so performance in one domain does not establish a fixed ordering of memory strategies. Experiment 6 shows that raw retrieval remains a competitive control in the public-flow setting.

Raw retrieval and semantic consolidation make different trade-offs. Raw examples retain labels and exceptions but consume context; consolidation can shorten that context while introducing unsupported conditions. We compare these alternatives empirically rather than assuming that abstraction improves predictions.

B. Localized Episodic Reflection

To mitigate the context window exhaustion and noise inherent in pure episodic retention, a subsequent wave of memory augmentation methodologies introduced the concept of Localized Episodic Reflection. Rather than storing raw experiences indefinitely, these approaches process experiences on a sample-by-sample basis, generating localized summaries, procedural critiques, or heuristic insights for individual episodes. While this significantly reduces the raw token count and highlights salient features, the architectural mechanisms used to generate these reflections vary, leading to distinct trade-offs in reliability and reasoning quality.

Early implementations of localized abstraction utilized Single-Step Direct Generation coupled with Episodic-Only Retrieval. For instance, Zhong et al. [16] proposed MemoryBank, which generates localized summaries of user interactions in a single forward pass to serve as long-term memory. While computationally inexpensive, this single-step architecture is highly susceptible to hallucination. Because the model is prompted to extract patterns without any intermediate reasoning or verification steps, it frequently generates flawed or overly simplistic heuristics that fail to capture the nuances of complex, multi-modal environments. Furthermore, by relying on Episodic-Only Retrieval during inference, the system treats these unverified summaries as standalone facts, propagating errors into the downstream reasoning tasks.

To address the limitations of single-step generation, more sophisticated frameworks adopted Multi-Step Linear Generation pipelines. A foundational work in this category is Reflexion [10], which allows agents to verbally reflect on individual past episodes through a sequential pipeline of evaluation and critique, thereby identifying mistakes to improve future performance. Similarly, Voyager [11] employs a linear pipeline to distill procedural skills from localized environmental interactions, while recent works like AMEM [17] and the framework proposed by Kim et al. [18] utilize multi-stage hierarchical summarization to structure localized memories.

Feedback and critique can improve subsequent behavior, but their availability does not by itself establish the validity of a stored heuristic. We distinguish task-level feedback from an explicit admission check against a labeled source episode.

Published systems differ in feedback, storage and retrieval, so the controlled adaptations used here should not be interpreted as complete reproductions of those systems.

These distinctions motivate an explicit per-episode admission check and a comparison of raw, abstract and hybrid context. We do not assume that other feedback-based systems are open-loop or that VLEG alone can bound errors. Their evaluation and memory mechanisms differ, and controlled adaptations need to be identified as such.

C. Global Semantic Consolidation

To overcome the limitations of localized reflection and pure episodic retention, a growing body of literature has focused on *Global Semantic Consolidation*. Methodologies in this category actively synthesize information across multiple disparate episodes to induce abstract, global rules or hypotheses. By distilling a noisy sea of unstructured experiences into a concise semantic memory, these approaches aim to minimize context window noise and maximize zero-shot generalization on novel tasks. The evolution of this paradigm can be traced through the increasing sophistication of its abstraction pipeline architectures.

The earliest approaches to global semantic consolidation relied on *Single-Step Direct Generation*. For instance, [13] proposed the IDEA framework, which prompts Large Language Models (LLMs) to perform induction, deduction, and abduction in a single forward pass to generate overarching hypotheses from text-based episodes. While computationally inexpensive and effective in low-complexity text domains, single-step generation is highly susceptible to hallucination. When applied to high-dimensional, multi-modal environments, these models frequently fail to autonomously distill accurate latent rules, as a single prompt is insufficient to disentangle complex, overlapping features without intermediate reasoning steps.

To mitigate the hallucination risks of single-step methods, subsequent research introduced *Multi-Step Linear Generation* pipelines. These frameworks divide the abstraction process into sequential, non-cyclic stages, such as hierarchical summarization or chain-of-thought factor extraction. Several works, including [19], [20], [21], successfully implemented multi-step linear pipelines to construct robust global rules. However, these systems predominantly utilize a *Semantic-Only Application* strategy during inference. By providing the inference model solely with the distilled global hypotheses, they entirely eliminate episodic noise but simultaneously deprive the model of concrete grounding examples. This creates a critical vulnerability: if the generated semantic rules are overly abstract, slightly inaccurate, or fail to capture edge cases, the model has no raw evidence to fall back on, potentially causing incorrect predictions. Addressing this integration flaw, [22] advanced the linear pipeline paradigm by incorporating *Hybrid Grounding*, allowing the model to reference both the linearly generated global rules and a subset of raw episodes at test time.

Despite the improvements in inference integration, multi-step linear pipelines still lack a mechanism to self-correct flawed abstractions during the memory generation phase itself. This limitation spurred the development of *Iterative Cyclic Refinement* architectures. Works such as [23] and [24] employ closed-loop generate-and-verify systems, where global hypotheses are iteratively proposed, tested against the episodic memory bank, and refined or pruned. While these cyclic frameworks maximize the semantic accuracy of the generated rules, they largely remain confined to Semantic-Only Application, again risking over-abstraction at inference time.

Global consolidation can preserve conditional rules when sufficient evidence and representational capacity are available. Our synthetic oracle has that property. The local-memory approach is evaluated as an alternative, with different admission and retrieval behavior; these experiments do not establish an inherent failure of global abstraction.

D. Hybrid Episodic-Semantic Grounding

The integration strategy employed during the inference phase dictates how a Large Language Model (LLM) balances high-fidelity contextual details against generalized reasoning heuristics. Recent comprehensive surveys on LLM memory mechanisms, such as those by Wu et al. [6] and Zhang et al. [7], emphasize that human-like cognitive architectures rely on a dual-pathway retrieval system. Relying solely on episodic retrieval risks overwhelming the context window with unstructured noise, whereas semantic-only application deprives the model of necessary grounding, frequently leading to hallucination via over-abstraction. Hybrid Episodic-Semantic Grounding addresses this dichotomy by conditioning the inference model simultaneously on raw episodic data and distilled semantic rules.

The state-of-the-art in this domain is exemplified by the Hypothesis-Driven Reasoning (HDR) framework [12]. HDR occupies a highly sophisticated and relatively sparse region of the methodological solution space, uniquely combining Global Semantic Consolidation, Iterative Cyclic Refinement, and Hybrid Grounding. During the inference phase on a novel task, HDR retrieves both the raw multi-modal episodes and the globally verified hypotheses. This dual-pathway integration allows the LLM to leverage abstract rules for global guidance while retaining the raw episodes for concrete visual or textual grounding, effectively bridging the gap between inductive rule learning and deductive application [12].

Despite its empirical success and robust architectural design, the reliance of HDR [12] on *Global* Semantic Consolidation introduces a critical vulnerability. By attempting to distill a single, unified set of global rules from the entire episodic memory bank, HDR implicitly assumes a homogeneous underlying data distribution. In highly heterogeneous or non-stationary environments, local episodes are frequently governed by mutually exclusive, context-dependent heuristics. Under such conditions, global rule extraction becomes potentially difficult under a restricted summary budget; the iterative refinement loop is forced

to either discard valid local rules (treating them as anomalies) or converge on overly generic, unhelpful hypotheses that fail to provide meaningful semantic guidance.

We examine the combination of per-episode admission checking and hybrid retrieval. The contribution is a controlled assessment of this combination and its limitations, rather than a claim that the combination resolves global abstraction in every heterogeneous setting.

III. The Proposed Method

A. Overview of Verified Localized Episodic Grounding (VLEG)

VLEG stores a labeled historical episode with an accepted local reflection. The implementation distinguishes generation, checking and inference roles. The following formulation describes their information access and stopping behavior; it does not guarantee correctness of the verifier or general validity of an accepted rule.

Let $\mathcal{E} = \{e_1, \ldots, e_n\}$ be historical episodes whose labels are available during consolidation. A pure episodic memory stores those labeled episodes directly. In the measured experiments, episodes are symbolic scene tuples. A network-flow implementation instead requires actual flow features and an independently labeled historical partition. Test labels are excluded from inference in either setting.

Conversely, global semantic consolidation [12] attempts to distill $\mathcal{E}$ into a compact set of universal rules or hypotheses, $\mathcal{H}$. This can be modeled as finding a global hypothesis set that maximizes the likelihood of the observed episodes:

$$\mathcal{H}^* = \arg\max_{\mathcal{H}} P(\mathcal{H} \mid e_1, e_2, \ldots, e_n) \tag{III.1}$$

However, in highly heterogeneous environments, the underlying data distribution is not governed by a single, unified set of rules. Instead, local episodes may be governed by mutually exclusive, context-dependent heuristics. In such cases, the optimization problem in Equation 1 becomes potentially difficult under a restricted summary budget, forcing the model to either over-abstract (losing critical conditional logic) or hallucinate false global consensus.

VLEG bridges this gap by shifting the abstraction target from the global distribution to the localized episode, while enforcing strict logical verification. Rather than attempting to find a universal $\mathcal{H}^*$, VLEG constructs a memory bank consisting of verified episodic-semantic tuples:

$$\mathcal{M}_{VLEG} = \{(e_i, r_i^*) \mid e_i \in \mathcal{E},\ \mathcal{V}(r_i^* \mid e_i) = 1\} \tag{III.2}$$

where r_i^* is a localized semantic reflection (a rule or heuristic) derived specifically from episode e_i, and $\mathcal{V}(\cdot \mid e_i)$ is a strict verification function that returns 1 when the implemented verifier accepts the candidate under its checking prompt; acceptance is a model judgment and can be wrong.

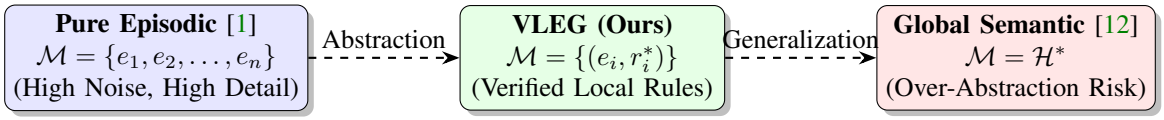


Fig. 1: The theoretical positioning of VLEG within the memory abstraction spectrum. VLEG bridges the gap between pure episodic retention and global semantic consolidation by pairing raw episodes with accepted local abstractions.

The stored tuple retains the source episode alongside its accepted reflection. This permits later comparison with the evidence, although the inference model may still follow an unsupported condition. The admission gate is only as reliable as the verifier for the specified error class.

B. Framework Components: Generator, Verifier, and Memory Bank

The implementation has a reflection generator, a verifier and a memory bank. Generation and checking use separate prompts and model calls. They use the same underlying model, so shared errors and dependent responses remain possible.

1) The Reflection Generator ($\mathcal{G}_\theta$): The Reflection Generator is an LLM parameterized by θ, tasked with distilling a high-dimensional raw multi-modal episode e_i into a localized, natural language abstraction r_i. Formally, at iteration $j = 0$, the generator samples an initial candidate reflection conditioned solely on the raw episode:

$$r_i^{(0)} \sim \mathcal{G}_\theta(\cdot \mid e_i) \tag{III.3}$$

In subsequent iterations ($j > 0$), if the initial reflection is deemed invalid, the generator is conditioned on the raw episode, its previous candidate reflection $r_i^{(j-1)}$, and a natural language critique $c^{(j-1)}$ provided by the verifier:

$$r_i^{(j)} \sim \mathcal{G}_\theta(\cdot \mid e_i, r_i^{(j-1)}, c^{(j-1)}) \tag{III.4}$$

While systems like Reflexion [10] terminate after generating $r_i^{(0)}$ and immediately commit it to memory, VLEG treats $\mathcal{G}_\theta$ strictly as a proposal distribution whose outputs are inherently untrusted until validated.

2) The Strict Verifier ($\mathcal{V}_\phi$): The verifier checks the candidate against one labeled historical episode. It can reject a mismatched label, an invented attribute or an uncheckable claim. This role is distinct from generation, but the two calls are not statistically independent.

The verifier outputs a tuple consisting of a binary validity score $v^{(j)} \in \{0, 1\}$ and a diagnostic critique $c^{(j)}$:

$$(v^{(j)}, c^{(j)}) = \mathcal{V}_\phi(e_i, r_i^{(j)}) \tag{III.5}$$

If $v^{(j)} = 1$, the candidate passes the implemented admission check. If $v^{(j)} = 0$, its critique is returned to the generator. Admission establishes only that this verifier accepted the reflection for this episode; it does not certify a universal rule or a lower false-positive rate than another system.

3) The Hybrid Memory Bank ($\mathcal{M}$): Once a reflection successfully passes the verifier (i.e., $v^{(j)} = 1$), it is denoted as the verified reflection r_i^*. The Hybrid Memory Bank $\mathcal{M}$ stores the paired tuple of the raw episode and its verified abstraction:

$$\mathcal{M} = \bigcup_{i=1}^{N} \left\{ (e_i, r_i^*) \mid \exists j \leq K \text{ s.t. } \mathcal{V}_\phi(e_i, r_i^{(j)})_v = 1 \right\} \tag{III.6}$$

where K is the maximum allowed refinement iterations. If an episode fails to yield a valid reflection within K iterations, it is discarded to preserve memory integrity.

The memory bank supports retrieval of both source evidence and a local reflection. This is the representation used in the hybrid modality; raw-only and abstract-only inputs remove one of its components for comparison.

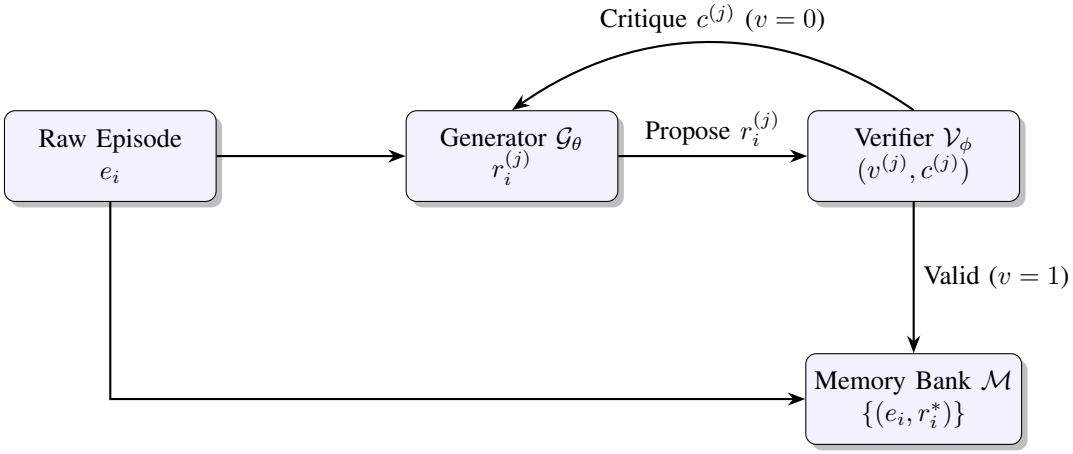


Fig. 2: Architectural flow of the VLEG framework components. Separate generator and verifier calls admit only candidates accepted by the implemented check to the Hybrid Memory Bank alongside their raw episodic counterparts.

C. Learning Mechanism: Iterative Cyclic Refinement

The refinement loop revises a rejected candidate using the verifier critique. It stops at the first accepted candidate or after K attempts. Reaching the budget without acceptance discards the episode from this reflection memory.

Formally, let $e_i \in \mathcal{E}$ denote a raw, high-dimensional multi-modal episode. The refinement process is governed by two interacting models: the Reflection Generator G_θ and the Strict Verifier V_ϕ. At the initial iteration ($j = 0$), the generator produces a candidate reflection conditioned solely on the raw episode:

$$r_i^{(0)} \sim G_\theta(\cdot \mid e_i) \tag{III.7}$$

Once a candidate reflection $r_i^{(j)}$ is generated, it is passed to the verifier V_ϕ. The verifier acts as a deterministic logical gate, evaluating the reflection strictly against the grounding evidence e_i. It outputs a tuple consisting of a binary validity score $v_i^{(j)} \in \{0, 1\}$ and a natural language critique $c_i^{(j)}$:

$$(v_i^{(j)}, c_i^{(j)}) = V_\phi(e_i, r_i^{(j)}) \tag{III.8}$$

Note that V_ϕ's only argument besides the candidate reflection is the single episode e_i it was generated from — by construction it has no access to any other episode. This can expose a reflection that misstates e_i's own attributes or label, but not sufficient to determine whether an attribute the reflection treats as necessary is actually causally relevant across episodes versus merely true of this one instance. We return to the practical consequence of this in the empirical caveat following the Hallucination Bounding remark below, and measure it directly in Experiment 5 (Section IV-F).

If $v_i^{(j)} = 1$, the reflection is deemed valid, the loop terminates, and the verified reflection $r_i^* = r_i^{(j)}$ is committed to the memory bank. If $v_i^{(j)} = 0$, the reflection contains a logical inconsistency or hallucination. The critique $c_i^{(j)}$ is then fed back into the generator to produce a refined reflection for the next iteration:

$$r_i^{(j+1)} \sim G_\theta(\cdot \mid e_i, r_i^{(j)}, c_i^{(j)}) \tag{III.9}$$

This process repeats until either $v_i^{(j)} = 1$ or a maximum iteration limit K is reached. If the limit K is exhausted without validation, the episode is deemed too ambiguous, and no reflection is stored, thereby prioritizing memory precision over recall.

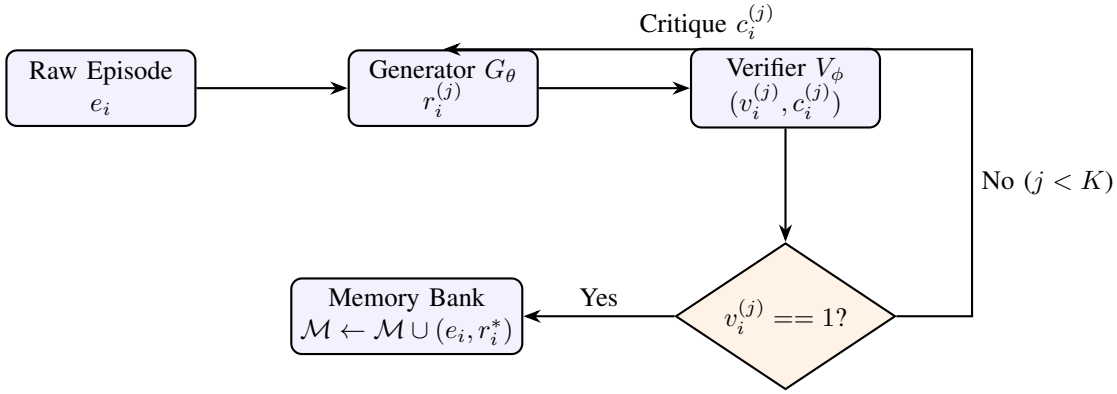


Fig. 3: State transition diagram of the Iterative Cyclic Refinement mechanism. The loop stores the first candidate accepted by the verifier to the semantic memory bank.

Remark III.1 (Scope of the admission check). Let A_j denote reaching attempt j, E_j an erroneous candidate at that attempt, and B_j its acceptance by the verifier. For $j = 1, \ldots, K$, suppose $\Pr(E_j \mid A_j) \le p_h$ and $\Pr(B_j \mid A_j, E_j) \le p_v$ for a specified error class. Since the loop stores the first accepted candidate,

$$\begin{aligned} \Pr(\text{erroneous commitment}) &\le \sum_{j=1}^{K} \Pr(A_j \cap E_j \cap B_j) \\ &\le \min\{1, K p_h p_v\}. \end{aligned} \tag{III.10}$$

This union bound needs no independence assumption. It does not imply that increasing K reduces erroneous commitments, nor does it bound error conditional on admission. A geometric power of p_v would require repeated acceptance checks on the same candidate, which the implemented loop does not perform. The probabilities are not estimated by the present experiments.

A single labeled episode can expose a contradiction or a fabricated attribute, but cannot establish which attributes cause a label or whether a heuristic transfers to other episodes. Experiment 5 measures this distinction in the symbolic setting; public-flow explanations require additional cross-episode or human assessment.

D. Inference Integration: Hybrid Episodic-Semantic Grounding

Once the memory bank $\mathcal{M} = \{(e_i, r_i^*)\}_{i=1}^N$ has been populated with accepted tuples through the iterative cyclic refinement process, the system must effectively leverage this knowledge to solve a novel target task T. Traditional retrieval-augmented generation (RAG) and standard in-context learning paradigms predominantly rely on episodic-only retrieval [2], [14], injecting raw, unstructured experiences directly into the context window. While this preserves high-fidelity details, it forces the reasoning module to perform ad-hoc abstraction at test time, which frequently leads to cognitive overload and degraded performance in complex, multi-modal environments.

To overcome this limitation, VLEG introduces a dual-pathway integration strategy termed *Hybrid Episodic-Semantic Grounding*. This approach simultaneously injects the raw multi-modal data e_i and its corresponding verified semantic rule r_i^* into the context window, aligning with advanced memory integration paradigms that advocate for structured, multi-level memory representations [6], [7].

1) Semantic-Driven Retrieval: Given a novel target task T, the system must first retrieve the top-k most relevant memory tuples. Because the raw episodes e_i are high-dimensional and often contain task-irrelevant noise, VLEG performs retrieval primarily in the semantic space defined by the verified reflections r_i^*.

Let $\phi(\cdot)$ denote a pre-trained dense embedding function. The relevance score $S(T, \mathcal{M}_i)$ for a given memory tuple $\mathcal{M}_i = (e_i, r_i^*)$ is computed using the cosine similarity between the embedding of the target task and the embedding of the verified reflection:

$$S(T, \mathcal{M}_i) = \frac{\phi(T) \cdot \phi(r_i^*)}{\|\phi(T)\| \|\phi(r_i^*)\|} \tag{III.11}$$

The system then selects the subset of k tuples that maximize this similarity score:

$$\mathcal{M}_{top} = \underset{\mathcal{S}\subset\mathcal{M},|\mathcal{S}|=k}{\arg\max} \sum_{(e_i,r_i^*)\in\mathcal{S}} S(T,(e_i,r_i^*)) \tag{III.12}$$

2) Hybrid Prompt Formulation: After retrieving $\mathcal{M}_{top}$, the inference module constructs a hybrid context window. The fundamental premise of hybrid grounding is that the verified reflection r_i^* provides the abstract, logical scaffold (the "what" and "why"), while the raw episode e_i provides the concrete, high-dimensional grounding (the "how"). This permits comparison with the source evidence, but does not guarantee that unsupported generalization is avoided.

The final inference prompt P is constructed by concatenating the target task T with the retrieved hybrid tuples, separated by structural delimiters:

$$P = \text{PromptBuilder}\left(\bigoplus_{(e_i,r_i^*)\in\mathcal{M}_{top}} \left[\text{Episode: } e_i \oplus \text{Verified Rule: } r_i^*\right] \oplus \text{Target: } T\right) \tag{III.13}$$

where $\oplus$ denotes the sequence concatenation operator. The reasoning LLM, parameterized by θ, then generates the final prediction $\hat{y}$ conditioned on this hybrid prompt:

$$\hat{y} = \text{LLM}_\theta(P) \tag{III.14}$$

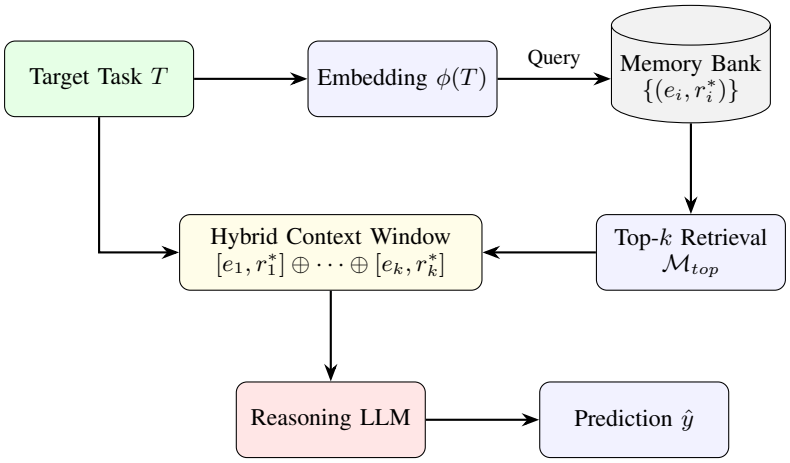


Fig. 4: The Inference Integration flow of VLEG. The target task queries the memory bank using semantic embeddings to retrieve the top-k tuples. Both the raw episodes and verified reflections are combined into a hybrid context window to ground the Reasoning LLM.

Hybrid context exposes both a reflection and its source evidence. It does not guarantee better predictions: irrelevant or over-specific reflections can also anchor the model on misleading attributes. Experiment 2 compares this modality with raw-only and abstract-only contexts under shared retrieval.

E. Final Algorithm and Execution Flow

The Verified Localized Episodic Grounding (VLEG) framework synthesizes the previously defined components—the Reflection Generator G_θ, the Strict Verifier V_ϕ, and the Hybrid Inference Module I_ψ—into a cohesive, end-to-end execution pipeline. This pipeline systematically bridges the gap between raw episodic retention [1] and abstract semantic reasoning [12] by ensuring that only accepted local heuristics are utilized during downstream inference.

The execution flow of VLEG is formally divided into two distinct phases: **Phase 1: Memory Consolidation** and **Phase 2: Hybrid Inference**.

During Phase 1, the system ingests a stream of raw, high-dimensional multi-modal episodes $\mathcal{E} = \{e_1, e_2, \ldots, e_n\}$. For each episode e_i, the Generator G_θ proposes an initial candidate reflection $r_i^{(0)}$. The Verifier V_ϕ then evaluates this candidate strictly against the grounding episode e_i, outputting a binary validity flag $v \in \{0,1\}$ and a natural language critique c. If $v = 0$ (indicating a hallucination or logical inconsistency), the critique c is fed back into G_θ to produce a refined reflection $r_i^{(1)}$. This iterative cyclic refinement continues until the reflection is validated ($v = 1$) or a predefined maximum iteration limit K is reached. If validated, the tuple (e_i, r_i^*) is committed to the Memory Bank $\mathcal{M}$. This strict gating mechanism mathematically bounds the propagation of unverified heuristics, directly addressing the vulnerabilities of single-step reflection pipelines [10].

During Phase 2, the system is presented with a novel target task T. A Retriever R computes the cosine similarity between the task embedding and the stored memory tuples, extracting the top-k most relevant memories $\mathcal{M}_{top} \subset \mathcal{M}$. The Inference

Module I_ψ is then prompted using Hybrid Episodic-Semantic Grounding, receiving both the raw episodes and their verified reflections to generate the final output. This dual-pathway integration aligns with advanced memory paradigms [6], making both representations available for the evaluated inference conditions.

The complete step-by-step procedure is formalized in Algorithm 1.

Algorithm 1 Verified Localized Episodic Grounding (VLEG)

Input: Episodic stream $\mathcal{E} = \{e_1, \ldots, e_n\}$, Target Task T, Max Iterations K, Retriever R

Initialize Memory Bank $\mathcal{M} \leftarrow \emptyset$
// Phase 1: Memory Consolidation
for each $e_i \in \mathcal{E}$ **do**
  $j \leftarrow 0$
  $c \leftarrow \emptyset$ ▷ Initialize empty critique
  $v \leftarrow 0$ ▷ Initialize validity flag to Invalid
  while $v = 0$ **and** $j < K$ **do**
    $r_i^{(j)} \leftarrow G_\theta(e_i, c)$ ▷ Generate or refine reflection
    $(v, c) \leftarrow V_\phi(e_i, r_i^{(j)})$ ▷ Verify reflection against raw episode
    $j \leftarrow j + 1$
  end while
  if $v = 1$ **then**
    $\mathcal{M} \leftarrow \mathcal{M} \cup \{(e_i, r_i^{(j-1)})\}$ ▷ Store verified tuple
  end if
end for
// Phase 2: Hybrid Inference
$\mathcal{M}_{top} \leftarrow R(\mathcal{M}, T, k)$ ▷ Retrieve top-k relevant tuples
Context $\leftarrow T \cup \mathcal{M}_{top}$ ▷ Construct hybrid prompt
Output $\leftarrow I_\psi$(Context) ▷ Generate final reasoning answer
return Output

To further illustrate the architectural data flow and the interaction between the distinct neural modules, Figure 5 provides a schematic representation of the VLEG execution pipeline. The diagram highlights the closed-loop nature of the memory consolidation phase and the dual-modality retrieval utilized during inference.

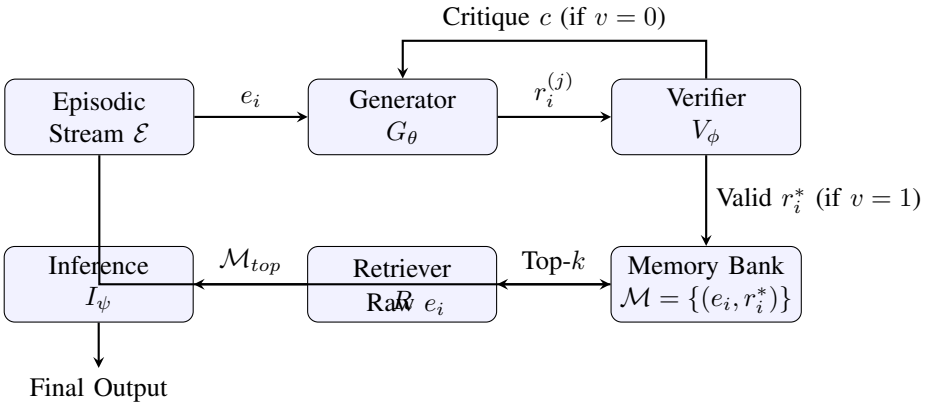


Fig. 5: Execution flow of the VLEG framework. The upper pathway illustrates the Iterative Cyclic Refinement loop (Phase 1), while the lower pathway demonstrates the Hybrid Episodic-Semantic Grounding retrieval (Phase 2).

The admission gate prevents storage when the verifier rejects a candidate. Its practical reliability depends on the verifier and the error class being checked. Retaining raw evidence permits downstream comparison with a reflection, but does not ensure that the model performs that comparison correctly.

IV. Experiments

A. Experimental Settings

Measured domain. Experiments 1–5 use the symbolic scene environment implemented in `core/task_env.py`. Its attributes are shape (circle/square), color (red/blue), texture (solid/striped), background (white/black), and a deterministically

assigned size distractor (small/large). Labels are ANOMALY and NORMAL. On a white background, the label is ANOMALY exactly when shape is circle and color is red; on a black background this condition is negated. Texture and size do not affect the oracle label. These are synthetic scene rules, not measured protocol, payload or network-zone features.

Evaluation population. Level 1 contains the eight shape–color–texture combinations on a white background. Level 2 contains all sixteen combinations across both backgrounds. For these levels, the same combinations form memory and evaluation queries, so accuracy measures recall/application to seen combinations. Level 3 withholds four combinations from memory and tests on all sixteen. Its overall accuracy mixes twelve seen and four unseen combinations; masked-only accuracy is the relevant measure of withheld-combination generalization. Seed pooling varies which combinations are withheld, not the underlying sixteen-element population. Repeated combinations across seeds must not be counted as independent traffic samples or evidence of zero-day detection.

Models and controls. The reported runs use Qwen2.5-7B-Instruct and Llama-3.3-70B-Instruct, with nomic-embed-text embeddings and temperature 0. The generator receives attributes without the historical label; the verifier receives the labeled source episode. Thus refinement uses supervised historical feedback. Accepted labeled episodes and reflections are available for retrieval, while target labels are used only for scoring. Generator and verifier are roles of the same model. Temperature 0 does not guarantee byte-identical results across inference-stack or execution changes.

Variables and baselines. The experiments vary consolidation strategy, grounding modality, model scale and $K \in \{1, 2, 3, 5\}$. Metrics include accuracy, admitted-memory coverage, diagnostic label consistency and unsupported attribution, and construction token counts. In Experiment 2, retrieval is shared across grounding modalities; Experiment 1's raw baseline retrieves raw-episode embeddings. The historical single-step comparator is an unverified reflection adaptation, not a full implementation of Reflexion. The current repository's Reflexion code instead includes multi-attempt environmental feedback; these old numbers have not been rerun with that changed implementation. Likewise, the global-hypothesis comparator is a controlled adaptation rather than a reproduction of every component of a published HDR system.

Provenance and evidence status. The numeric results and figures below are retained from the supplied manuscript. The current repair requires genuine JSON booleans in both verifier pipelines; old results predate this repair and have not been revalidated against archived verifier replies. Their association with the exact model weights, prompts and code revision must be resolved before submission. The manuscript reports cross-scale checks for all five experiments, with per-experiment seed counts stated below; model-scale comparisons are descriptive and do not isolate parameter count from model family. The repository also contains a separate rendered-CLEVR pipeline, which is not a public network-flow validation and is not merged into these numeric tables. Hardware, quantization, offload state and model digests require run-specific documentation; the manuscript's earlier blanket hardware attribution is not used for all runs.

B. Experiment 1: Controlled Memory-Strategy Comparison

Experiment 1 compares VLEG, global-hypothesis consolidation, unverified single-step reflection, and raw in-context retrieval on the three symbolic task levels. The comparisons characterize this small environment and these adaptations; they do not establish superiority over published systems on their original benchmarks.

Within each model-scale comparison the same underlying model and temperature are used. VLEG uses $K = 3$. Level 1 has eight queries and Level 2 sixteen. Level 3 is pooled over three masking seeds, giving 48 evaluations of sixteen recurring combinations. Repeat runs matched reported accuracies in the checked conditions, which is an empirical repeatability observation rather than a determinism guarantee.

The reported 7B accuracies are ICL 100.0%, 93.8% and 87.5% at Levels 1–3, versus VLEG 75.0%, 87.5% and 75.0%. Global-hypothesis consolidation recovers the uniform-background rule at Level 1 (100.0%), but has pooled accuracy 70.8% at Level 3. The historical single-step reflection comparator scores 25.0%, 50.0% and 50.0%; the inspected reflections all predict ANOMALY, making its accuracy equal to the corresponding anomaly base rate. These values describe the historical adaptation, not the changed multi-attempt feedback baseline now present in the repository.

At 7B, raw retrieval performs best in this environment. This limits the evidence for an advantage from consolidation at that scale. Experiment 2 isolates grounding content more directly, and the limitations below explain why this globally expressible oracle does not establish a general weakness of global consolidation.

Cross-scale check. The reported 70B check evaluates Levels 1 and 2 once each and Level 3 across three masking seeds. The latest archived, feedback-baseline revision gives Level-3 pooled accuracies of VLEG 91.67% (44/48), raw retrieval 89.58% (43/48), global hypothesis 83.33% (40/48), and feedback-based Reflexion 81.25% (39/48). VLEG exceeds raw retrieval in two seeds but is lower in the third. These replace the earlier 93.75% VLEG and single-step-baseline numbers; the feedback-based comparator is not the same condition as the 7B first-proposal comparator. This reverses the descriptive ordering at 7B. Because the model family also changes and the evaluation population is small, the difference should not be attributed to parameter count alone.

C. Experiment 2: Component Isolation (Grounding Modality Ablation)

Experiment 2 isolates which retrieved content is shown to the inference model: the raw episode, the accepted reflection, or both. The same retrieved tuples are used across conditions to reduce retrieval-quality confounding.

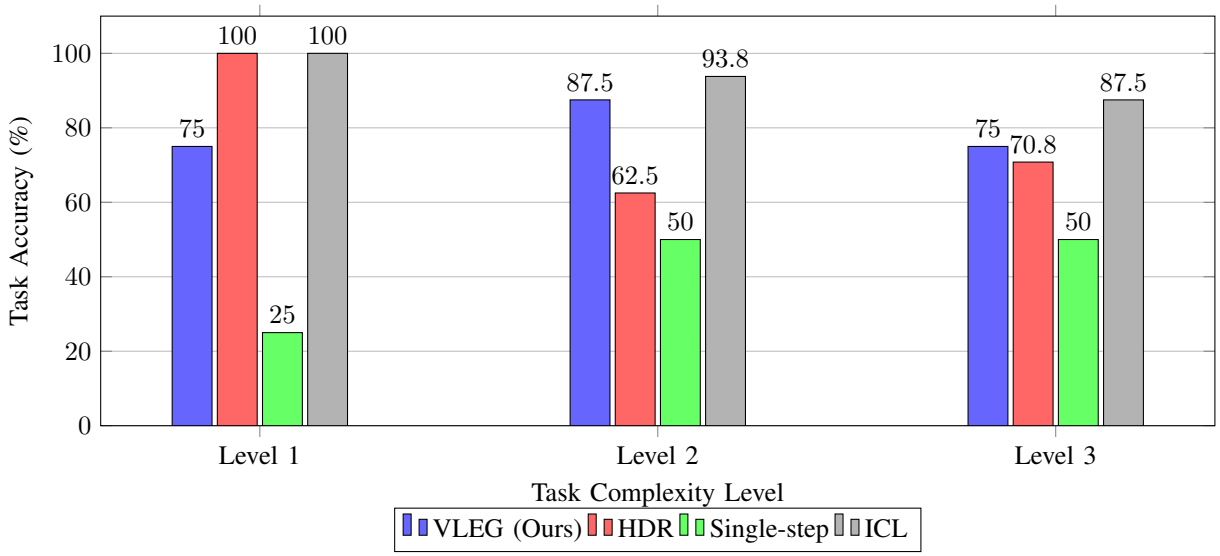


Fig. 6: Measured Task Accuracy (%) across three levels of task complexity (reduced-order reimplementation, Qwen2.5-7B-Instruct; Level 3 pooled across 3 masking seeds, $n = 48$). ICL leads at every level; HDR is bimodal (perfect at Level 1's non-conditional rule, weak once the rule becomes background-conditional); Single-step tracks each level's ANOMALY base rate rather than discriminating between scenes.

To achieve this, we evaluate the framework under three distinct ablation conditions. The first condition is *Hybrid Grounding (VLEG)*, which represents the complete proposed methodology where the reasoning module is conditioned on both the retrieved raw episodes e_i and their corresponding verified reflections r_i^*. The second condition is *Abstract-Only Grounding*, where the reasoning module is provided solely with the verified reflections r_i^*, completely omitting the raw multi-modal episodic data. The third condition is *Raw-Only Grounding*, which serves as a baseline mirroring standard In-Context Learning (ICL) [1], where the model receives only the raw episodes e_i without any distilled semantic guidance.

To isolate the effect of grounding *content* from retrieval quality, all three conditions retrieve the same top-k memory tuples (via cosine similarity on the verified reflection's embedding, as VLEG's memory bank is built the same way regardless of which fields are shown at inference); only which fields of each retrieved tuple are shown to the reasoning LLM differs between conditions.

The measured results, shown in Figure 7, do not match the hypothesized ordering. On Level 2 (no masking), Hybrid Grounding is indeed the best condition (87.5% vs. 81.2% for both alternatives). But on Level 3 — the condition this experiment was specifically designed to stress-test — the ordering inverts: Hybrid Grounding is the *worst* of the three conditions, both overall (73.8% pooled across 5 masking seeds, $n = 80$, vs. 78.8% for Abstract-Only and 75.0% for Raw-Only) and specifically on the masked combinations that never appeared during memory formation (60.0% pooled, $n = 20$, vs. 70.0% for both alternatives). Hybrid never outperformed both alternatives on Level 3 in any of the 5 seeds tested. We connect this to Experiment 5 (Section IV-F): verified reflections frequently cite causally irrelevant attributes as if they were determinative (see the empirical caveat in Section III), and pairing a raw episode with such a reflection appears to anchor the reasoning model on the reflection's stated — sometimes spurious — condition rather than letting it reason more freely from the raw examples, which is more costly precisely when generalization to an unseen combination is required.

At 7B, abstract-only context exceeds hybrid context by 5.0 percentage points overall and 10.0 points on the masked subset. The result does not support a claim that hybrid context is necessary for this task. The reported 70B check below has a different overall ordering.

Cross-scale check. We repeated this experiment with Llama-3.3-70B-Instruct across the same 5 masking seeds. Unlike Experiments 1, 3, and 5, where the 7B result either reversed cleanly (Experiments 1, 3) or stayed exactly as limited (Experiment 5), this one produced a genuinely mixed picture. Pooled over 5 seeds ($n = 80$ overall, $n = 20$ masked-only): overall accuracy, Hybrid (95.0%) now ties Raw-Only (95.0%) for best, with Abstract-Only worst (91.25%) — the opposite of Abstract-Only's 7B standing. But on the masked-combo subset specifically, Abstract-Only remains the best modality (85.0% vs. 80.0% for both alternatives) — the same direction as the 7B finding, just with a narrower margin (5 points versus 10). Per-seed masked accuracy for Abstract-Only was noisy ($100\%, 75\%, 50\%, 100\%, 100\%$ across the 5 seeds), with one seed accounting for most of its overall shortfall. We do not read this as either a clean confirmation or a clean reversal of the 7B finding: the specific claim that Hybrid is the worst modality overall does not hold at 70B, but the narrower claim that Abstract-Only retrieval has some edge specifically on unseen, masked combinations persists in direction across both scales, just less dramatically.

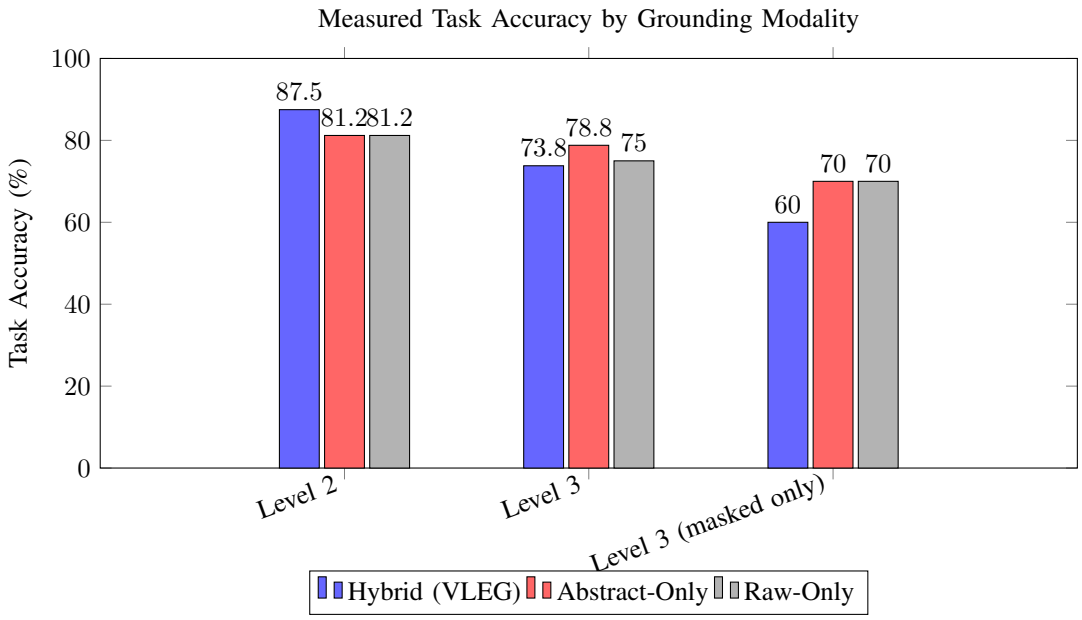


Fig. 7: Measured Task Accuracy across grounding modalities (reduced-order reimplementation, Qwen2.5-7B-Instruct). Level 3 figures are pooled across 5 masking seeds ($n = 80$ overall, $n = 20$ for the masked-only subset). Hybrid Grounding wins on Level 2 but is the worst condition on Level 3, especially on the masked combinations it was intended to help with — the opposite of the original hypothesis.

D. Experiment 3: Iterative Refinement Ablation

Experiment 3 tests whether increasing the refinement budget recovers rejected episodes and changes prediction accuracy. It does not assume that iteration or verification is necessary in every task.

Both $K = 1$ and $K = 3$ run the same generate-and-verify procedure. At $K = 1$, the one rejected proposal leads to discarding the episode; at $K = 3$, the generator can revise after critique. This is a refinement-budget ablation, not a verified-versus-unverified comparison. Both use the same model and hybrid inference.

Across five masking seeds, $K = 3$ achieves 75.0% overall accuracy $(60/80)$ versus 63.75% $(51/80)$ for $K = 1$, a difference of 11.25 percentage points. Masked-only accuracy is 60.0% $(12/20)$ versus 50.0% $(10/20)$. The masked-only difference is zero in three seeds and positive in two. These counts are repeated evaluations under different memory masks, not eighty independent scenes.

The mechanism behind the gain is memory bank *coverage*, not reasoning quality per reflection: across the 5 seeds, $K = 1$'s memory bank retained only 42–58% of training episodes (an episode whose first-guess reflection is rejected is discarded outright, since j is never $< K{=}1$ again), while $K = 3$'s retained 83–100%, because the critique from a rejected first attempt typically reveals the correct label and lets the generator correct itself on the second or third try. Verification at $K = 3$ is thus better understood here as *recovering episodes that a single unlucky guess would otherwise lose*, not as improving the causal quality of the reflections that do get committed — see Experiment 5 (Section IV-F) for a direct measurement showing no such quality improvement.

Cross-scale check. At 70B, the reported mean overall difference is 26.2 percentage points (standard deviation 8.1) across the same five seeds, and $K = 3$ admits all twelve historical episodes in each seed. The results are consistent with coverage recovery through label-informed critique; they do not show that the accepted explanations identify causal attributes more reliably.

E. Experiment 4: Parameter Sensitivity and Overhead Analysis

The primary objective of Experiment 4 is to systematically analyze the convergence behavior of the VLEG verification loop and to quantify the associated computational trade-offs. While the iterative generate-and-verify mechanism is designed to strictly filter out hallucinated heuristics, each refinement cycle inherently incurs additional computational costs during the memory consolidation phase. Therefore, it is critical to evaluate how the strictness of the verification process impacts both the downstream reasoning performance and the overall token efficiency of the system.

To conduct this sensitivity analysis, we manipulate the maximum refinement iterations parameter, K, across a discrete set of values: $K \in \{1, 2, 3, 5\}$. The condition $K = 1$ effectively disables the iterative refinement loop, reducing the system to a single-step reflection generator akin to standard localized reflection baselines [10]. For each configuration of K, we evaluate the model on both Level 2 and Level 3 of the Experience-Transfer task. We record the resulting Task Accuracy (%) alongside the

Computational Overhead, which is measured as the total number of prompt and completion tokens consumed per consolidated episodic memory.

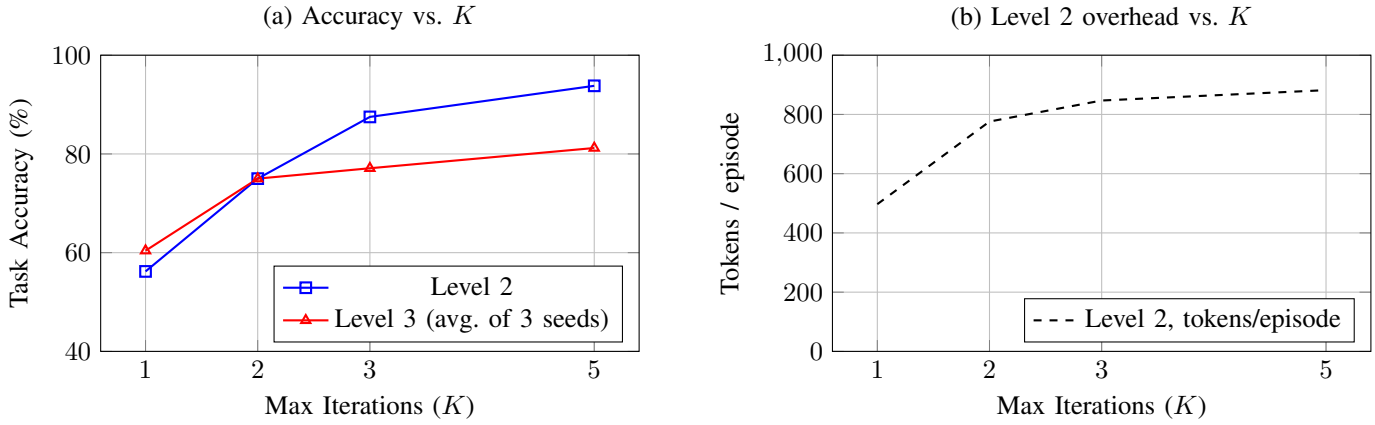


Fig. 8: Measured parameter sensitivity of the maximum refinement iterations K (reduced-order reimplementation, Qwen2.5-7B-Instruct). (a) Task Accuracy does *not* plateau at $K = 3$: Level 2 (a single deterministic run, confirmed identical on repeat) keeps improving through $K = 5$; Level 3 (averaged over 3 masking seeds) plateaus in only 1 of 3 seeds individually. (b) Token overhead per consolidated episode is concave, not linear: most of the cost increase happens between K=1 and K=2, with sharply diminishing marginal cost for extending the budget further, since only a minority of episodes ever use the additional iterations.

The measured results, shown in Figure 8, contradict both claims originally hypothesized for this experiment. First, Task Accuracy does not plateau around $K = 3$. On Level 2 — a single run, verified deterministic by an exact repeat — accuracy keeps climbing substantially from K=3 (87.5%) to K=5 (93.8%), a further +6.3 point gain with no sign of leveling off in this range. On Level 3, we tested 3 independent masking seeds: only one of them (seed = 42) shows the exact plateau signature (identical bank composition, token count, and accuracy at K=3 and K=5); the other two show K=5 either matching or beating K=3 by a further +6–12.5 points. We do not have a reliable stopping criterion at $K = 3$ in this environment — it happens to be sufficient for some random draws of the masked subset and insufficient for others.

Second, Computational Overhead does not scale linearly with K. It is concave: the token cost per consolidated episode rises sharply from K=1 to K=2 (+279 tokens/episode on Level 2) and then far more slowly for each additional unit of K budget (+71 tokens/episode from K=2 to K=3, +35 from K=3 to K=5 despite that step covering two extra iterations, not one). This pattern was consistent across all seeds tested on both levels. The mechanism is straightforward: most episodes converge within 1–2 iterations regardless of the ceiling K; raising the ceiling only adds cost for the small minority of episodes that are still unresolved at iteration 3, so the marginal cost of a higher K budget shrinks as K grows even as the marginal *benefit* (per the accuracy panel) does not reliably shrink in the same way. Practitioners should not treat $K = 3$ as a validated efficiency-optimal cutoff on the basis of this experiment; whether a higher K is worth its (modest, sub-linear) additional cost depends on how much the specific downstream task benefits from the extra recovered episodes, which Experiment 3 (Section IV-D) suggests is real but seed-dependent in magnitude.

Cross-scale check. At 70B, the reported Level-3 sweep uses three masking seeds. In each, $K = 2$, $K = 3$ and $K = 5$ yield the same admitted memory, token consumption and accuracy. No episode needs a third attempt in these runs. The observed plateau is therefore at $K = 2$ in this model and task, while the 7B sweep is less uniform. This does not establish a universal stopping budget.

Concurrent GPU workloads make the affected timing measurements unsuitable for a clean overhead comparison. One reported repeat at Level 2, $K = 3$, changes the token count from 12,541 to 12,625 while preserving accuracy. Accordingly, temperature 0 is recorded as a decoding setting, and repeatability and hardware contention must be assessed empirically for each run.

F. Experiment 5: Diagnostic Assessment of Memory Quality

Experiment 5 separates source-label consistency from unsupported attribution. These diagnostics complement downstream accuracy and test what a per-episode admission check actually filters in this symbolic domain.

We report admitted-memory coverage, source-label consistency and a keyword-based unsupported-attribution diagnostic. Coverage is the proportion of episodes admitted within $K = 3$; it is not meaningful as a verification metric for an always-commit comparator. Label consistency is scored against the known synthetic label. The attribution flag detects references to texture or size, which are irrelevant to the oracle. It can flag a mere mention as well as an asserted condition, so it is a coarse diagnostic. Earlier joint LLM judgments are excluded from the reported table because they conflated these axes.

Table I reports 100% source-label consistency among admitted VLEG reflections at both levels, versus 41.7–50.0% for the unverified single-step comparator. Conditional on label correctness, unsupported-attribution rates are 86.7% versus 87.5% at Level 2 and 83.3% versus 80.0% at Level 3. These small-sample rates show no clear verification-specific reduction. Source-label consistency is an observed diagnostic outcome, not a tautological guarantee of an imperfect model-based verifier.

TABLE I: Diagnostic Assessment of Memory Quality (Experiment 5, reduced-order reimplementation, Qwen2.5-7B-Instruct). Label correctness and spurious-attribution rate are computed deterministically against ground truth, not via LLM judgment. "Spurious (of correct)" = % of label-correct reflections that still cite a causally irrelevant attribute (texture/size) as determinative.

Level	Strategy	Validity	Label-correct	Spurious (of correct)
Level 2	Single-Step	N/A	50.0% (8/16)	87.5% (7/8)
	VLEG (K=3)	**93.8%**	**100.0% (15/15)**	86.7% (13/15)
Level 3	Single-Step	N/A	41.7% (5/12)	80.0% (4/5)
	VLEG (K=3)	**100.0%**	**100.0% (12/12)**	83.3% (10/12)

In this diagnostic, the admitted reflections match their source labels, but frequently cite attributes that do not affect the synthetic oracle. For example, asserting that striped texture is necessary cannot be validated from a single striped instance. Cross-episode contrast is needed to distinguish an observed property from a general condition. A keyword flag is a coarse attribution diagnostic, not a causal explanation assessment for natural traffic.

Cross-scale check. At 70B, reconstructing the source order prescribed by the experiment script gives admitted-reflection label consistency of 100.0%, versus 50.0% for first proposals. This pairing is not independently attested by a checkpoint. Unconditional texture/size keyword rates are equal for both strategies: 12.5% (2/16) at Level 2 and 16.7% (2/12) at Level 3, compared with approximately 80–87% at 7B. The reduction occurs for both strategies, so it is not evidence of a verification-specific benefit. Given the small counts, equality should not be described as proof of statistical equivalence. Cross-scale grounding results are reported separately in Experiment 2.

G. Experiment 6: Public-Flow Benchmark Validation

This experiment tests whether local verification and hybrid memory improve classification of independent public flow records. It uses UNSW-NB15, a captured testbed benchmark containing normal and generated attack traffic [25]. It is not a production deployment or zero-day detection study. The completed archive contains 500 pilot predictions and 25,000 formal predictions; an offline audit reproduces the metrics, conditional bootstrap intervals and construction-cost totals from the stored records.

Data and partition audit. The source partitions contain 175,341 historical and 82,332 test records. The frozen feature representation uses nineteen fields: duration, protocol, service, state, directional packet and byte counts, rate, TTL, load, inter-packet timing, jitter and mean packet size. IPs, timestamps, row identifiers, attack names and target labels are excluded from model inputs and retrieval keys. The public mirror is pinned to a commit; its file names are reversed relative to the official partition sizes, so role mapping is recorded explicitly and checked by row count, size, MD5 and SHA256. Bytes agree with an independent research deposit, although the original author-hosted files were not directly accessible.

Identical selected-feature vectors sometimes have conflicting labels. We exclude every row in such groups rather than choosing a label: 6,896 historical rows in 273 groups and 2,912 test rows in 105 groups. We also remove 79,746 and 33,161 remaining same-label duplicate rows, respectively, leaving 88,699 historical and 46,259 test fingerprints. A further 1,044 test fingerprints overlap the cleaned historical source and are excluded. Thus the evaluated population is deduplicated and unambiguous in this representation; removing these cases can change difficulty and class balance, so these results do not estimate performance on the original flow-frequency distribution. Session-level independence cannot be established because this partition lacks session-group metadata.

Frozen evaluation and implementation. Each of five memory seeds (20261020–20261024) selects sixteen benign and sixteen malicious historical examples. They share the same 1,000 test fingerprints: 663 benign and 337 malicious. The 100 pilot queries do not overlap these formal queries. We use Llama-3.3-70B (70.6B parameters, Q4_K_M), nomic-embed-text (137M, F16), temperature 0, $K = 3$, $k = 3$, and a 32,768-token chat-context ceiling. The run records exact model digests, source/code hashes and Python 3.13.13; the server has two A800 80GB GPUs. Generator and verifier are separate calls to the same model. Historical labels are available to verification and the raw baseline, but target labels are used only after inference for scoring. The generator's first proposal receives only historical flow features.

The five conditions are raw flow retrieval, an unverified first reflection, VLEG, raw flow retrieval restricted to VLEG's admitted examples, and a single global summary of the labeled historical memory. All local conditions share a label-free raw-feature embedding retrieval pass and the same selected top-three candidates. A rejected candidate is dropped without replacement. The raw-on-admitted control therefore removes reflection content while retaining the same admission mask as VLEG. This shared raw retrieval deliberately differs from the reflection-driven formulation in Section III and tests consolidation/admission under controlled retrieval; it does not evaluate every component of the full formulation. Global-summary inference sees a corpus

TABLE II: Public-flow performance, mean ± sample standard deviation across five historical-memory seeds (percent). All seeds use the same 1,000 test fingerprints. Invalid outputs count as errors.

Condition	Accuracy	Macro-F1	Attack recall	False positive rate
Raw flow retrieval	65.08 ± 2.93	63.92 ± 2.54	71.39 ± 8.44	38.13 ± 6.85
Unverified single-step	61.38 ± 1.08	42.66 ± 1.35	6.17 ± 1.43	9.89 ± 2.03
VLEG	59.74 ± 2.03	59.68 ± 2.03	77.74 ± 5.69	48.78 ± 1.67
Raw on admitted flows	64.86 ± 2.96	63.70 ± 2.68	71.16 ± 8.71	38.34 ± 6.61
Global summary	50.22 ± 12.45	47.60 ± 14.13	87.12 ± 22.27	68.54 ± 24.20

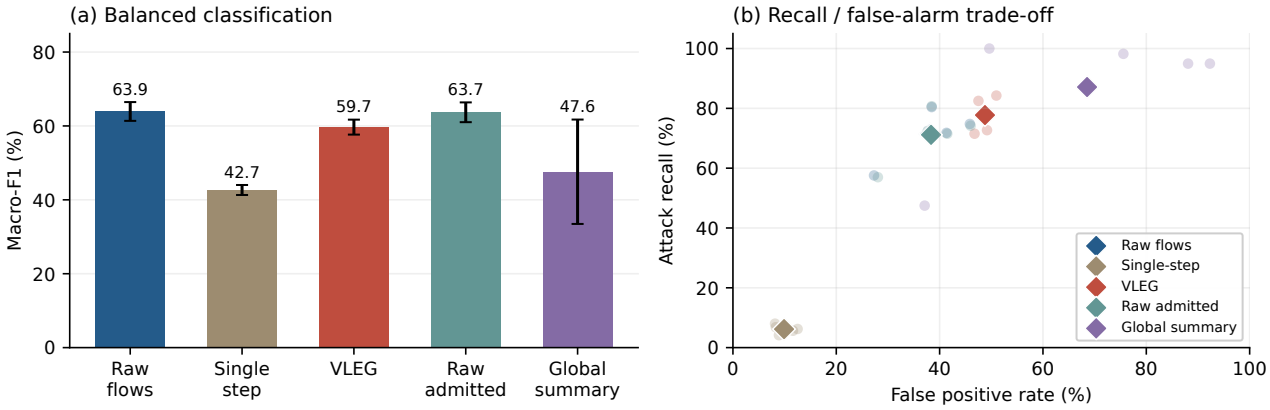


Fig. 9: Public-flow results across five memory seeds. Left: mean Macro-F1 with sample-standard-deviation bars. Right: attack recall versus false positive rate; faint points are individual seeds and large markers are means. The larger recall of VLEG is accompanied by more false alarms.

summary rather than the same top-three examples, and is a memory-architecture comparison rather than an equal-evidence modality ablation. These are controlled adaptations, not complete published Reflexion/HDR reproductions.

Performance and trade-off. Table II and Figure 9 show that raw retrieval has the highest mean Macro-F1 among the evaluated conditions (63.92%), while VLEG reaches 59.68%. VLEG increases attack recall from 71.39% to 77.74%, but increases the false positive rate from 38.13% to 48.78%. Its accuracy falls from 65.08% to 59.74%. A post hoc always-benign reference has accuracy 66.30%, Macro-F1 39.87% and zero attack recall; this illustrates why accuracy alone is insufficient here. None of these conditions establishes deployment-quality intrusion detection, and no comparison against a trained conventional intrusion classifier is made.

Verification is beneficial relative to the unverified first-reflection condition: Macro-F1 improves by 17.02 percentage points, and attack recall rises from 6.17% to 77.74%. However, high admitted-memory coverage (98.75%, 158/160 historical episodes) does not establish reliable cross-flow explanations. Raw-on-admitted retrieval nearly matches the raw baseline (63.70% versus 63.92% Macro-F1), whereas adding the accepted reflections reduces its mean score to 59.68%. These comparisons are consistent with reflection content changing the decision trade-off; they do not establish a specific anchoring or causal-attribution mechanism. A subsequent deterministic explanation screen finds source-label agreement in 158/158 admitted VLEG reflections versus 84/160 first proposals. Explicit conjunctive conditions can be evaluated in 157/158 VLEG reflections; all 157 hold on their source flow, but 113 have at least one contrary-label match among the other archived historical draws. The remaining expression uses ambiguous $\gg$ syntax and is unresolved. The screen uses 160 unique historical flows and no test queries. It checks literal predicates, not natural-language causal validity, and the historical pool is neither independent nor representative. A human review with field-semantic judgments and adjudication remains incomplete.

Paired uncertainty and repeated-query dependence. A paired bootstrap resamples the same 1,000 fingerprints synchronously across methods and all five memory seeds (1,000 replicates). The mean VLEG-minus-raw Macro-F1 difference is -4.24 percentage points, with a conditional 95% interval of $[-5.91, -2.53]$; VLEG is lower in four seeds and slightly higher in one. VLEG-minus-single-step is $+17.02$ points, interval $[13.71, 20.39]$. VLEG-minus-raw-on-admitted is -4.02 points, interval $[-5.79, -2.28]$. These are unadjusted descriptive intervals conditional on the fixed memory draws and an assumption of independent fingerprints; they do not quantify uncertainty over new networks, sessions, model weights or memory-seed populations. Five repeated evaluations do not create 5,000 independent test flows. Per-attack recalls are archived, but rare categories include only one Backdoor, one Worms and three Shellcode cases; their percentages cannot support precise category-specific conclusions.

Costs and repeatability. Table III separates one-time memory construction from query chat time. VLEG uses a mean 40,094 construction tokens and 419.43 seconds, versus 9,405 tokens and 126.54 seconds for first-proposal reflections. Raw

TABLE III: Observed public-flow costs, means across five memory seeds. Construction tokens include input and output. Chat latency excludes shared retrieval and is confounded by fixed method order/cache reuse.

Condition	Construction calls	Construction tokens	Construction (s)	Query mean (s)	Query p95 (s)
Raw flow retrieval	0	0	0	2.12	2.67
Unverified single-step	32.0	9,405	126.54	2.23	2.75
VLEG	120.4	40,094	419.43	1.74	2.68
Raw on admitted flows	120.4	40,094	419.43	0.95	1.19
Global summary	1.0	5,947	23.21	1.21	1.30

storage needs no construction chat calls; embedding cost is separate and embedding tokens are not included. VLEG query input averages 909 tokens versus 780 for raw retrieval. Methods run in the fixed order raw, single-step, VLEG, raw-on-admitted, global summary. Consequently, prompt caching and order confound latency comparisons: the shorter observed VLEG chat time is not evidence of a causal speed advantage. Moreover, among 4,968 raw/raw-on-admitted pairs with identical reconstructed prompts, 24 predictions differ despite temperature 0. This variation limits interpretation of the small raw-versus-admission-control gap. There are 29 invalid single-step outputs and 30 invalid VLEG outputs across 5,000 predictions each; all other conditions have zero invalid outputs. False positive rate counts explicit malicious predictions on benign flows, so invalid-output counts and benign recall must be considered alongside it.

The admission-control condition shares VLEG's construction process and its cost; it does not obtain the mask for free. The completed experiment therefore supports a bounded result: checking first reflections improves on an unverified adaptation, but accepted hybrid reflections do not improve balanced classification over raw retrieval on this cleaned public-flow population.

H. Limitations

Experiments 1–5 use small symbolic populations, while Experiment 6 evaluates a cleaned public-flow benchmark. The latter is a controlled adaptation with shared raw retrieval, not a production deployment or a complete validation of reflection-driven retrieval. Levels 1 and 2 evaluate seen combinations; Level 3 combines seen and withheld combinations. Their accuracy must not be interpreted as held-out network generalization. The flow experiment is restricted by deduplication/conflict exclusion, missing session metadata, one model and test partition, fixed method order and incomplete human explanation audits.

The synthetic rule is expressible as a single background-conditioned statement. It tests conditional rule learning, not an impossibility of global abstraction. The 7B and 70B comparisons also change model family and cannot isolate scale causally. Observed rank reversals and equal diagnostic rates are descriptive findings in these samples, not general superiority or equivalence claims.

Verification sees a known historical label and tests a candidate against one source episode. It can reject inconsistencies detected by the model, but agreement does not establish causal attribution or cross-episode validity. Reflection diagnostics based on known synthetic attributes cannot be transferred unchanged to natural traffic, where ground-truth explanations are generally unavailable.

Finally, the 7B figures are linked to the archived initial reimplementation snapshot, while updated 70B comparisons use the latest archived feedback-baseline revision. Stored item-level accuracies were independently recounted, but these historical files omit exact model digests and original verifier replies. Boolean type validation was also repaired in the oracle-audit parser; without original replies its historical impact cannot be quantified. Experiments 1–5 were not rerun, while Experiment 6 is independently audited against its archived predictions and frozen source. Context limits, quantization, model digests, CPU/GPU offloading and concurrent workloads require run-specific recording. The completed public-flow run records source/model/code hashes and hardware, but does not resolve every limitation of the historical experiments.

V. Conclusions

VLEG checks local reflections before memory admission and combines accepted reflections with source episodes during inference. In the reported symbolic experiments, refinement increases memory coverage and admitted reflections are more consistent with historical labels. Downstream benefits depend on the model and task: raw retrieval leads at 7B, while the latest archived 70B comparison gives a one-correct-evaluation pooled advantage to VLEG over raw retrieval. Hybrid grounding retains a disadvantage on some withheld combinations, and per-episode verification does not show a separate benefit for unsupported attribution.

The public-flow experiment strengthens this bounded interpretation. At 70B, VLEG improves Macro-F1 over unverified first reflections but is lower than raw retrieval, while increasing both attack recall and false alarms. High memory admission coverage therefore does not establish reliable cross-flow reasoning. Further work should test calibration and cross-episode validation on new, independently reserved data, complete explanation audits, and control timing/repeatability before making deployment claims. Historical symbolic files have now been inventoried against immutable 7B and 70B Git snapshots, and 48 stored item-level accuracy records reproduce their summaries. The latest feedback-baseline results replace stale 70B comparisons. Git

provenance cannot recover missing model digests or raw verifier replies, so corrected-parser reruns and human review remain required.

REFERENCES

[1] T. B. Brown, B. Mann, N. Ryder, M. Subbiah, J. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, S. Agarwal, A. Herbert-Voss, G. Krueger, T. J. Henighan, R. Child, A. Ramesh, D. M. Ziegler, J. Wu, C. Winter, C. Hesse, M. Chen, E. Sigler, M. Litwin, S. Gray, B. Chess, J. Clark, C. Berner, S. McCandlish, A. Radford, I. Sutskever, and D. Amodei. Language Models are Few-Shot Learners. *ArXiv*, abs/2005.14165, 2020.

[2] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, and D. Kiela. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. In H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, and H. Lin, editors, *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474. Curran Associates, Inc., 2020.

[3] C. Packer, S. Wooders, K. Lin, V. Fang, S. G. Patil, I. Stoica, and J. E. Gonzalez. MemGPT: Towards LLMs as Operating Systems. *ArXiv*, abs/2310.08560, 2023.

[4] W. Wang, L. Dong, H. Cheng, X. Liu, X. Yan, J. Gao, and F. Wei. Augmenting Language Models with Long-Term Memory. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

[5] Y. Wang, Y. Gao, X. Chen, H. Jiang, S. Li, J. Yang, Q. Yin, Z. Li, X. Li, B. Yin, J. Shang, and J. McAuley. MEMORYLLM: Towards Self-Updatable Large Language Models. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235, pages 50453–50466, 2024.

[6] Y. Wu, S. Liang, C. Zhang, Y. Wang, Y. Zhang, H. Guo, R. Tang, and Y. Liu. From Human Memory to AI Memory: A Survey on Memory Mechanisms in the Era of LLMs. *ArXiv*, abs/2504.15965, 2025.

[7] Z. Zhang, X. Bo, C. Ma, R. Li, X. Chen, Q. Dai, J. Zhu, Z. Dong, and J.-R. Wen. A Survey on the Memory Mechanism of Large Language Model based Agents. *ArXiv*, abs/2404.13501, 2024.

[8] C. An, J. Zhang, M. Zhong, L. Li, S. Gong, Y. Luo, J. Xu, and L. Kong. Why Does the Effective Context Length of LLMs Fall Short? *ArXiv*, abs/2410.18745, 2024.

[9] N. F. Liu, K. Lin, J. Hewitt, A. Paranjape, M. Bevilacqua, F. Petroni, and P. Liang. Lost in the Middle: How Language Models Use Long Contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173, 2024.

[10] N. Shinn, F. Cassano, A. Gopinath, K. R. Narasimhan, and S. Yao. Reflexion: language agents with verbal reinforcement learning. In *Thirty-seventh Conference on Neural Information Processing Systems*, 2023.

[11] G. Wang, Y. Xie, Y. Jiang, A. Mandlekar, C. Xiao, Y. Zhu, L. Fan, and A. Anandkumar. Voyager: An Open-Ended Embodied Agent with Large Language Models. *Transactions on Machine Learning Research*, 2024.

[12] Aakash Kumar Agarwal and Moyuru Yamada. Hypothesis-Driven Reasoning for Large Language Models. In *Proceedings of the Fortieth AAAI Conference on Artificial Intelligence (AAAI-26)*, pages 1686–1693, 2026.

[13] K. He, M. Zhang, S. Yan, P. Wu, and Z. Chen. IDEA: Enhancing the Rule Learning Ability of Large Language Model Agent through Induction, Deduction, and Abduction. In *Annual Meeting of the Association for Computational Linguistics*, 2024.

[14] Y. Zhang, S. Feng, and C. Tan. Active Example Selection for In-Context Learning. *ArXiv*, abs/2211.04486, 2022.

[15] K. Purohit, V. V, S. Bhattacharya, and A. Anand. Sample Efficient Demonstration Selection for In-Context Learning. In *Forty-second International Conference on Machine Learning*, 2025.

[16] W. Zhong, L. Guo, Q. Gao, H. Ye, and Y. Wang. MemoryBank: Enhancing Large Language Models with Long-Term Memory. *Proceedings of the AAAI Conference on Artificial Intelligence*, 38(17):19724–19731, 2024.

[17] W. Xu, Z. Liang, K. Mei, H. Gao, J. Tan, and Y. Zhang. A-MEM: Agentic Memory for LLM Agents. *ArXiv*, abs/2502.12110, 2025.

[18] S. H. Kim, K. T. Ong, T. Kwon, N. Kim, K. Ka, S. Bae, Y. Jo, S. Hwang, D. Lee, and J. Yeo. Towards Lifelong Dialogue Agents via Timeline-based Memory Management. In *North American Chapter of the Association for Computational Linguistics*, 2024.

[19] K. il Lee, H. Koh, D. Lee, S. Yoon, M. Kim, and K. Jung. Generating Diverse Hypotheses for Inductive Reasoning. *ArXiv*, abs/2412.13422, 2024.

[20] T. Kim, M. Cochez, V. François-Lavet, M. Neerincx, and P. Vossen. A machine with short-term, episodic, and semantic memory systems. In *Proceedings of the Thirty-Seventh AAAI Conference on Artificial Intelligence and Thirty-Fifth Conference on Innovative Applications of Artificial Intelligence and Thirteenth Symposium on Educational Advances in Artificial Intelligence, AAAI'23/IAAI'23/EAAI'23*, 2023.

[21] J. Li and J. Li. Memory, Consciousness and Large Language Model. *ArXiv*, abs/2401.02509, 2024.

[22] Y. Wang and X. Chen. MIRIX: Multi-Agent Memory System for LLM-Based Agents. *ArXiv*, abs/2507.07957, 2025.

[23] L. Qiu, L. Jiang, X. Lu, M. Sclar, V. Pyatkin, C. Bhagavatula, B. Wang, Y. Kim, Y. Choi, N. Dziri, and X. Ren. Phenomenal Yet Puzzling: Testing Inductive Reasoning Capabilities of Language Models with Hypothesis Refinement. In B. Kim, Y. Yue, S. Chaudhuri, K. Fragkiadaki, M. Khan, and Y. Sun, editors, *International Conference on Representation Learning*, volume 2024, pages 1452–1481, 2024.

[24] Y. Zhou, H. Liu, T. Srivastava, H. Mei, and C. Tan. Hypothesis Generation with Large Language Models. In *Proceedings of EMNLP Workshop of NLP for Science*, 2024.

[25] Nour Moustafa and Jill Slay. UNSW-NB15: A comprehensive data set for network intrusion detection systems (UNSW-NB15 network data set). In *Military Communications and Information Systems Conference (MilCIS)*, 2015.